

Robertson Derivation Document — Reading Guide

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README: How to Read the Robertson Mathematical Derivation & Numerical Analysis Document

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Project Overview

This repository hosts the mathematical derivation and numerical analysis documentation for the Robertson chemical kinetics problem. The Robertson problem is a classic stiff system of ordinary differential equations (ODEs) describing the evolution of three reactant concentrations over time. Its three rate constants span roughly nine orders of magnitude, causing fast-decaying and slow-evolving modes to coexist within the same system. This coexistence of multiple time scales is the very source of stiffness, and also the central question of this project: Explicit methods are constrained by the stability requirements of the fast mode — can implicit methods achieve comparable accuracy at lower computational cost?

This document is the deliverable of Team Member 1, covering all mathematical derivations and theoretical analysis required by the first 12 items of the project brief. Every theoretical conclusion is paired with a concrete numerical experiment specification for the coding team to implement and verify.

Repository Files

File	Description
Robertson_数学推导与数值分析文档.pdf	Core deliverable: Team Member 1's mathematical derivation document, 27 pages, 12 chapters
README.pdf	This file: a reading guide to help team members quickly understand the document's structure and usage

Document Organization

Every chapter in the document strictly follows a four-part structure, ensuring that each theoretical conclusion has a corresponding numerical experiment:

1. Theory: Mathematical derivations starting from first principles — rather than citing textbook formulas, results are rebuilt from fundamental tools such as Taylor expansions.

2. Prediction: Concrete, testable predictions derived from the theory, intended for verification through numerical experiments.
3. Required Experiments: A clear specification of the numerical experiments the coding team must implement, including function interfaces and output requirements.
4. Explanation Notes: How to interpret the experimental results, common pitfalls, and connections to subsequent chapters.

Reading tip: If you only care about "what to implement", jump straight to the "Required Experiments" subsection of each chapter. If you need to understand "why it is designed this way", read the "Theory" subsection in order.

Document Structure & Chapter Overview

The document consists of 12 chapters, organized in a logical progression: "Model formulation → Method derivation → Stability analysis → Stiffness analysis → Experiment design". The table below lists each chapter's topic and core content:

Chapter	Title	Core Content
§1	Robertson Model & Basic Properties	ODE system definition, conservation law derivation, positivity, role of diagnostics
§2	Three Numerical Methods: Formulas & Theoretical Order	Taylor-based derivations and convergence orders of Explicit Euler, RK4, and Implicit Euler
§3	Newton Iteration for Implicit Euler	Residual function, Newton formula derivation, linear solve vs. explicit inversion, tolerance effects
§4	Convergence Theory & Experimental Criteria	Observed order computation, asymptotic regime conditions, error definitions and code specs
§5	Absolute Stability of the Three Methods	Stability region derivations, A-stability, importance for Robertson
§6	Robertson Jacobian Derivation	Component-wise representation of the RHS, term-by-term partial derivatives, full Jacobian matrix
§7	Structural Zero Eigenvalue & Initial Degeneracy	Mechanism by which the conservation law forces a zero eigenvalue, implications for stiffness analysis
§8	Stiffness Ratio & Reduced System	Stiffness ratio definition, 2x2 reduced Jacobian, numerical zero-eigenvalue threshold
§9	Stability Limit of Explicit Euler on Robertson	From absolute stability to step-size upper bound, locally frozen Jacobian estimate
§10	Unified Diagnostic Quantities	Four core diagnostics, project-level pass/fail rules, common misjudgments
§11	Matched-Error Comparison	Why equal step size is unfair, the matched-error scheme, cost metrics
§12	Newton Tolerance Sweep	Competition between two error sources, sensitive diagnostic quantity, expected results

Suggested Reading Paths

Team members in different roles can follow the paths below to retrieve the information they need most efficiently:

Coding Team (Recommended Path)

1. Read §1 first to understand the Robertson ODE system and the two diagnostics: conservation and positivity.
2. Read §2 to learn the formulas and theoretical convergence orders of the three methods, and to settle on a unified interface design.
3. Read §3 to grasp the Newton iteration implementation details for each Implicit Euler step (linear solve, tolerance settings).
4. Read §4 to clarify the experimental criteria for convergence tests (four halving step sizes, observed-order computation).
5. Read §6 and §8 to implement the Jacobian function and stiffness ratio computation.
6. Read §10 to implement the four core diagnostics and the pass/fail rules.
7. Read §11 and §12 to design the matched-error comparison and Newton tolerance sweep experiments.

Shortcut: The "Required Experiments" subsection of each chapter already lists function interfaces and output requirements — it can serve directly as an implementation spec.

Theory Team (Recommended Path)

1. Follow §1 → §2 → §5 to understand the connection between the model, methods, and stability theory.
2. Read §6 → §7 → §8 to understand the Jacobian structure, the origin of the zero eigenvalue, and stiffness analysis.
3. Read §9 for the quantitative estimate of how Explicit Euler is constrained by stiffness.
4. Read §4 and §11 for the methodology of convergence theory and fair comparison.

Quick Overview (Recommended Path)

1. Read §1.1 and §1.2 for the background of the Robertson problem and the conservation law.

2. Read the §2.4 comparison table of the three methods for a quick grasp of their characteristics.
3. Read §5.5 to understand why A-stability is crucial for stiff systems.
4. Read §10.2 and §10.3 for the four core diagnostics and the pass/fail rules.

Prerequisites

The following background is recommended for reading this document:

- Ordinary Differential Equations: initial value problems, existence and uniqueness, Lipschitz condition.
- Numerical Analysis Fundamentals: Taylor expansion, local truncation error vs. global error, explicit vs. implicit methods.
- Linear Algebra: matrix eigenvalues, Jacobian matrix, LU factorization and linear solves.
- Numerical Stability Concepts: absolute stability region, A-stability, and the basic meaning of stiffness.
- Newton's Method: linearization of nonlinear equations, convergence conditions.

Relationship Between This Document and the Code

This document covers the mathematical derivation and theoretical analysis only — it does not contain code. The "Required Experiments" subsection of each chapter provides the coding team with a clear implementation spec, including:

- Function names and interfaces to implement (e.g., `robertson_jacobian(t, y)`, `Newton solver`, etc.).
- Output quantities to record (e.g., residual, iteration count, converged/failed status, max error).
- Experiment design parameters (e.g., step-size sweep range, tolerance sweep sequence, four halving step sizes).
- Verification criteria (e.g., whether the observed order matches the theoretical order, whether conservation and positivity hold).

If the coding team has questions about a derivation during implementation, they can trace back to the "Theory" subsection of the corresponding chapter for the full derivation. The "Explanation Notes" subsection also highlights common pitfalls and things to watch for — it is recommended to consult it when analyzing experimental results.

Key Terminology Quick Reference

Term	Meaning	Ref.
Stiffness	Coexistence of multiple time scales forcing explicit methods to use tiny steps	§1, §8
Conservation law	Total mole fraction of the three species remains constant	§1.2
Positivity	Chemical concentration components must be non-negative	§1.3
A-stability	Absolute stability region contains the entire left half-plane	§5.5
Absolute stability region	Range of step sizes for which the method is stable for a given eigenvalue	§5.1
Observed order	Actual convergence order recovered from experimental data by regression	§4.2
Asymptotic regime	Region where step size is small enough, inside the stability region, and the leading error term dominates	§4.4
Structural zero eigenvalue	A zero eigenvalue forced in the Jacobian by the conservation law	§7
Stiffness ratio	Ratio of the largest to smallest non-zero eigenvalue magnitudes, measuring stiffness	§8.1
Reduced Jacobian	Reducing the 3x3 system to 2x2 via the conservation law	§8.2
Matched-error comparison	A fair scheme: drive each method to the same accuracy, then compare cost	§11
Newton tolerance sweep	Vary the Newton iteration tolerance and observe the effect on accuracy and efficiency	§12

Contact & Feedback

If you have questions about the document, find a derivation error, or would like a theoretical detail expanded, please submit a GitHub Issue and reference the relevant chapter number (e.g., §6.2) for quick triage.